

Pre-Requisites

- Familiarity with Python programming and data structures (lists, dictionaries).
- Fundamental understanding of NLP basics (tokenisation, stopwords removal).
- Basic knowledge of machine learning concepts (features, models).
- Prior exposure to Bag-of-Words and TF-IDF representations.

Session Summary

This 180-minute session moves beyond surface-level text processing to focus on *meaning*. We introduce ambiguity and lexical relations, then work through four cornerstone tasks in semantic analysis: Word Sense Disambiguation (WSD), Named Entity Recognition (NER), Semantic Role Labelling (SRL), and Coreference Resolution. We also discuss modelling approaches for sequence labelling, highlighting the role of Conditional Random Fields (CRFs).

Learning Objectives / Outcomes

- Explain lexical ambiguity (polysemy vs. homonymy) and why it matters for semantics.
- Differentiate the goals of WSD, NER, SRL, and Coreference Resolution.
- Apply the IOB schema to annotate entities for NER.
- Analyse SRL outputs to identify who-did-what-to-whom (agent, theme, recipient).
- Use WordNet to explore semantic relations (hyponymy, meronymy).
- Describe the high-level function of CRFs for sequence labelling in NER.

Brief Agenda

- **Foundations of Meaning (00:15–00:50)**
Lexical semantics, ambiguity (polysemy vs. homonymy), and core lexical relations (with WordNet demo).
- **Representation & WSD (00:50–01:20)**
Network vs. distributional views; Lesk algorithm demo; discussion of WSD challenges.
- **NER & IOB Tagging (01:30–02:00)**
Core entity types (PER/ORG/LOC), IOB labelling scheme; hands-on annotation.
- **Models for NER: CRFs (02:00–02:25)**
How CRFs model dependencies; feature examples (capitalisation, POS); quick visualisation.
- **SRL & Coreference (02:25–02:55)**
Extracting predicate–argument structure; linking mentions across sentences; ambiguity pitfalls.
- **Recap & Q&A (02:55–03:05)**
Summary of tasks and when to use them; open questions.

Detailed Agenda

Table 1: Detailed 180-Minute Agenda for M2: Core Tasks in Semantic Processing

Time (min)	Session Focus and Activities
00:00–00:10	Session Kickoff & Goals
	Expectations, outcomes, and how semantic tasks fit into the NLP stack.
00:10–00:15	Welcome & Agenda
	Motivation for semantic processing and overview of the flow.
00:15–00:50	Block 1: Foundations of Meaning
	What is lexical semantics? Polysemy vs. homonymy with examples. Explore lexical relations; WordNet demo to visualise hyponymy/meronymy. Short Q&A.
00:50–01:20	Block 2: Representation & WSD
	Representing meaning: network (WordNet) vs. distributional (VSM). Task: WSD using the Lesk algorithm (worked example). Q&A on failure cases and context sensitivity.
01:20–01:30	Break
	Short break.
01:30–02:00	Block 3: NER & IOB Tagging
	Define NER and common tag sets (PER/ORG/LOC). Apply IOB labelling to annotate entity boundaries; quick peer review of tags. Q&A on edge cases.
02:00–02:25	Block 4: Models for NER (CRF)
	Introduce Conditional Random Fields for sequence labelling. Show feature design (capitalisation, POS, affixes) and discuss why CRFs capture label dependencies. Compare briefly with alternatives.
02:25–02:55	Block 5: SRL & Coreference
	SRL: identify predicate and arguments; read model output to extract who-did-what-to-whom. Coreference: cluster mentions (e.g., “she”, names) across sentences; discuss ambiguity and discourse context.
02:55–03:05	Recap & Final Q&A
	Summarise the four tasks, typical pipelines, and when to use each. Open questions.

Conclusions

- Semantic understanding in NLP is a multi-step pipeline: disambiguate, extract entities & roles, and link references.
- CRFs are a strong baseline for structured prediction in tasks like NER due to label-dependency modelling.
- SRL and coreference add structure and context critical for downstream applications (question answering, dialogue).

Additional Readings

- Explore WordNet to build intuition for lexical hierarchies and relations.

- Review distributional semantics (PMI, vector space models) to contrast with network views.
- Optional refresher: Bag-of-Words and TF-IDF as foundations for text representation.